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MétaCan: study protocol

An open, provenance-tracked, coverage-audited map of Canadian metaresearch

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Status: preregistration draft. To be registered on OSF **before the frame is built and before any retrieval is run**. Registration date, OpenAlex snapshot version, and Érudit harvest date will be stamped into RELEASE.md at v1.0 and cannot be revised afterwards.

1. Rationale

Mapping a research community with bibliographic metadata invites a specific failure: the map records what the retrieval could see, then quietly presents itself as a map of the field. The gap between the two is rarely measured, and where it is measured it is often large. In earlier work I benchmarked automated publication linkage for funded trials against a manual reference standard built by two independent reviewers. Conditional on a candidate reaching the matcher, it performed well (sensitivity 92.2%, PPV 94.3%); unconditionally, its sensitivity was **44.5%**. The failure was in retrieval, and it was invisible to every internal quality check.

MétaCan is built so that the same failure, if it occurs, is measured rather than concealed. Two commitments follow, and they drive every design decision below:

1. **Provenance.** Each record states which route(s) retrieved it and which sense of “Canadian” it satisfies.
2. **A measured gap.** A human audit samples the records the pipeline did *not* retrieve, so that retrieval sensitivity is estimated rather than assumed.

2. Preliminary work (complete, and reported in the proposal)

The pilot analyses are reproducible via `make pilot` and re-derivable offline from archived responses via `make pilot-offline`. The authoritative list, with every value, is **pilot/results/FINDINGS.md**, which is *generated* from `pilot/results/findings.json`; neither the list nor its COUNT is restated here, because a hand-maintained copy of a generated table is a stale table waiting to happen. This protocol previously carried a seven-row version of it, which was already wrong by the time anyone read it, and then a hand-typed count, which went stale one sentence away from this warning.

The five that determine the design:

#	Result	Design consequence
12	Scored against the rubric, the best topic route retrieves 12% of Canadian meta-research (95% CI 5.6 to 21.6), at 60% precision. It misses 66 of 75 . OpenAlex files a work by what it is <i>about</i> , so meta-research about cardiology reads as cardiology.	The field is invisible to topic retrieval precisely because it is about other fields. Retrieval is demoted to provenance; screening over a Canadian frame is the gate.
10	Swap which model is “the screener” and the base rate moves from 1.06% to 2.37% (design-weighted, same 1,290 records): 37,032 vs 83,022 works. The published binomial CI contains neither.	Machine agreement is a process metric , not accuracy. The screener-swap range, not the binomial interval, is the honest uncertainty. The human audit is the study.
11	31.5% of the frame has no abstract, and the screen finds 0.78% meta-research there against 1.55% where one exists ($p = 0.023$; robust to adjustment for year and language). An earlier version of this table also claimed the blindness was differential (T2 losing 3.6x against T1’s 1.4x). That cell holds four works and the interaction is not significant ($p = 0.141$). WITHDRAWN (DEVIATIONS.md D6).	A third of the frame is screened on its title alone and the screen finds half as much there. The audit stratifies on abstract availability. It is not evidence of differential blindness by tradition, and this protocol no longer says it is.
14	A simple random sample of the screened-out mass cannot measure screening sensitivity : 600 records return an expected 0.4 misses; twenty would need 2,009 coder-hours against 65 budgeted.	The two-phase audit as first specified was impossible. See §6.0 and §6.1.

#	Result	Design consequence
16	Haiku lands near Sonnet's base rate (1.27% vs 1.06%, 98.1% agreement) but their in-scope sets overlap 16% unweighted, 10% design-weighted; of Sonnet's 58 positives Haiku confirms 12 . And agents of ONE model on ONE prompt disagree beyond chance after conditioning on what they were shown (CMH $p = 0.0056$, replicated at 0.015), with the agents' ordering flipping between arms.	Rate agreement is not set agreement, and no machine pass measures this field. The machine screen is demoted to a stratifier for the audit: design weights stay unbiased under a noisy stratifier, only efficiency suffers, and power is computed at Haiku-realistic concentration. Machine tiers are released marked provisional; every reported prevalence is design-weighted human coding.
D1	The pilot screen did not obey this rubric. It was sent 6 of the 8 fields §5 mandates: venue, OpenAlex topic and field, Canadian affiliations and funders were all withheld. Every pilot number came from a title-and-abstract screen.	The most serious defect in the project. The full screen sends the rubric's complete payload and reports the difference as a finding. See DEVIATIONS.md D1.

Also established: 0 of 4,516 OpenAlex topics name the field; 64% of works in the topic space have no raw affiliation string; Érudit matches 0 OpenAlex sources but exposes 379 harvestable OAI-PMH sets; reproducibility alone returns 43,392 Canadian works at 0.8% on-topic; NSERC carries 9.2x SSHRC's linked works; naive capture-recapture implies Canada produces 59% of world metaresearch against an observed 1.9%, and is **abandoned**; the OpenAlex API is now metered, so the frame is read from a pinned S3 snapshot; and screening the frame with the full rubric twice costs **~\$4,767** on the Batch API (§7). An earlier version of that estimate charged the rubric once per WORK rather than once per CALL, reported \$24,379, and used it to justify a cheap TRIAGE in front of the screen. The triage is a retrieval step, in a study whose central finding is that retrieval destroys these maps. With the arithmetic corrected it saves \$110, so **it is deleted**: every work in the frame is read against the full rubric. (DEVIATIONS.md D7.)

3. Estimand (prespecified)

The field boundary is contestable. That is a reason to decide in advance and be explicit, not a reason to defer.

Primary estimand: Canadian-produced metaresearch. A work is in scope if it is classified T1 or T2 (§5.2) **and** satisfies at least one of: - **CA-AFF**: ≥ 1 authorship with an institutional affiliation whose country is Canada, **or** whose OpenAlex-parsed authorship country is Canada even where the institution is unresolved (the second clause exists because finding 2 shows 64% of the topic space carries no resolvable affiliation); - **CA-FUND**: ≥ 1 funder whose OpenAlex country_code is Canada. This is an **external criterion, not a curated list**: an earlier version of this clause named four funders (CIHR, NSERC, SSHRC, CFI), and that list failed twice: three of its four hardcoded IDs pointed at

other funders, and even corrected it missed Canada Research Chairs (42,824 works), the Government of Canada, NRC, Mitacs, Genome Canada, and every provincial agency. OpenAlex knows **2,182** Canadian funders; `R/pin_entities.R` pins them, and the pinned list ships with the data.

The two components are **reported separately as well as jointly**, because pilot finding 6 shows CA-FUND is skewed against the social sciences, and a pooled figure would hide that.

Secondary estimand: metaresearch about the Canadian research system, irrespective of author location, identified by screening rather than by keyword.

Descriptive attributes, NOT inclusion criteria. Recorded on every record, never used to admit one:
 - **CA-VENUE**: published in a Canadian venue (incl. Érudit-hosted); - **CA-PERSON**: an author with a known prior Canadian affiliation, at any time.

Rationale for the exclusion: a work by someone who once held a Canadian post, written abroad, with no Canadian funding, affiliation, subject matter, or venue, is not self-evidently Canadian metaresearch. Nor is a globally generic paper that happens to appear in a Canadian journal. These are interesting covariates and poor gatekeepers.

Prespecified alternative rules (to test the consequences of the primary decision, not to replace it): - **ALT-1 (strict)**: CA-AFF only. - **ALT-2 (broad)**: $\text{primary} \cup \text{CA-VENUE} \cup \text{CA-PERSON}$.

Only these two alternatives will be reported. We will not enumerate all combinations of the five flags: that is researcher degrees of freedom wearing the costume of a sensitivity analysis.

4. Frame (F)

F is defined and frozen before screening, and every claim in this study is bounded to F.

The design inverts the usual one. The obvious approach retrieves what *looks like* metaresearch and then asks whether it is Canadian. That makes the field boundary a property of the retrieval strategy, and it is why such maps cannot be audited: a lexicon cannot show you what it never surfaced, so there is nothing left to measure the miss against.

Here the frame is **all Canadian research**: an external, checkable criterion that owes nothing to our notion of metaresearch. Field membership then becomes a **classification** question over an enumerable universe, not a **retrieval** question over the literature. That is what makes “what did we miss?” answerable at all, and it is what produced the base rate (§2, finding 9) that replaces the failed coverage estimator.

F IS NO LONGER A HYPOTHESIS. IT IS BUILT. All 482 partitions of the pinned snapshot were streamed and filtered: **F holds 4,299,418 works**, each appearing exactly once (finding 23). For most of this project’s life F was “about 3.5M”, an **extrapolation from a single partition** that sat hardcoded in six scripts; **the estimate was 18% low**, and every cost, audit-power and field-size figure computed against it has moved. `R/frame_size.R` now reads the size from the artifact the harvest writes, so no script can hardcode it again.

Sources that CONTRIBUTE RECORDS to F. - A pinned OpenAlex snapshot (release stamped; not the live API, which is metered and mutable. See finding 8: 1,000 credits per ~11h window, and 8.2 days per pass. A study behind a meter that resets every eleven hours cannot be re-run by a reviewer). - A pinned Érudit OAI-PMH harvest (<https://oai.erudit.org/oai/>, 379 sets; harvest date stamped).

Sources that DO NOT contribute records, and must not. Retraction Watch, ClinicalTrials.gov, and the CIHR project database are used, and **none of them admits a work to F**. A retracted cardiology

paper is retracted *cardiology*; a trial registration is not a publication; a grant is not a work. Admitting them would let an interesting signal masquerade as the estimand, which is the error this protocol exists to prevent. They enter as: - **Record attributes.** Retraction Watch supplies the post-publication state (finding 17): OpenAlex's `is_retracted` is a *boolean over a four-value space* (retraction, expression of concern, correction, reinstatement), so it expresses one and silently reports the rest as false. RW is joined by DOI and its state and reasons ship with F. - **Reference standards.** ClinicalTrials.gov knows a Canadian trial happened *independently of any pipeline*, so it cannot be wrong in the pipeline's favour. It is the only reference standard here **not made of machine labels** (finding 21), and §6.1 uses it for known-item recall alongside the venue set. - **Enrichment.** PubMed / Europe PMC / Crossref supply abstracts OpenAlex lacks (finding 19). **Preprints are already in F** and need no ingest: measured against bioRxiv and medRxiv's own API, OpenAlex indexes 99.6% of what the servers hold (finding 20).

Window. Publication years 2000–2025 inclusive.

Membership. A record enters F if it carries **any identifiable Canadian signal**: CA-AFF · CA-FUND · Canada named in title, abstract or keywords (EN/FR) · published in a Canadian venue · is an Érudit record. Every record stores **which routes admitted it**. **1,565,226 works (36.4% of F) carry no Canadian affiliation at all**: an affiliation-only frame would hold 2,734,192 works and would never see them. That is the empirical case for the union of routes, and for provenance on every record.

Limitations, stated up front. - A work whose Canadian link is invisible to the metadata (no affiliation, no funder, no textual mention, no Canadian venue) cannot enter *any* Canadian frame, by any method. That is a hard boundary of the data, not of this design. - Scholarship indexed by neither OpenAlex nor Érudit is **not estimated** here, and no claim about it will be made. - **Both clauses of the primary estimand rest on sparse metadata.** 64% of the metaresearch topic space carries no raw affiliation string (finding 2), and **71.2% of F carries no funder metadata at all** (finding 18). CA-FUND cannot admit a work whose funder OpenAlex never recorded; that is a hard ceiling on the route, and it is why the audit must sample the works **no route reached**.

5. Screening (retrieval is demoted to provenance)

Every record in F is **classified**, not retrieved. Retrieval routes still run (candidate topics and venues; bilingual lexical and semantic similarity to the seed corpus; citation neighbours of curated seeds), and every record stores the set of routes that surfaced it. But those routes are **provenance, not the gate**: they answer “how would a conventional search have found this?”, which is exactly the question the coverage audit needs, and they are never allowed to decide membership.

Route overlap and incremental yield are reported descriptively. They are **not** used to estimate the unseen population (§6.3).

5.2 Inclusion tiers

- **T1 (core):** the work's object of study is research itself: its methods, reporting, reproducibility, integrity, evaluation, funding, workforce, communication, or infrastructure.
- **T2 (adjacent):** empirically or conceptually grounded work on science as a social system (STS, library and information science, research infrastructure studies) that is not framed as metaresearch but studies it.
- **T3 (contextual):** policy documents, editorials, commentary, announcements. **Mapped and released, never pooled into the analytic corpus.**
- **Excluded:** everything else.

5.3 Genre label

empirical · conceptual · editorial/commentary · policy · infrastructure/announcement · other. Assigned by the classifier because OpenAlex's type is unreliable and this field is unusually commentary-heavy.

5.4 Classifier

Multilingual sentence embeddings (cross-lingual by construction) over title + abstract, plus an LLM screening pass with **published prompts** and temperature 0. The decision **threshold is locked before evaluation** and recorded in the protocol registration.

The classifier plays no part in constructing the reference standard against which it is judged (§6.2). No LLM output is shown to the human coders.

6. Validation: two-phase stratified probability audit

The core of the study, and the thing the field usually skips.

6.0 The design this replaces, and why it could not work

An earlier version of this protocol said: partition F into screened-in and screened-out, draw a probability sample from both, human-code them, and estimate sensitivity with design weights. That is the textbook two-phase audit and it is **arithmetically incapable of doing the job here**. Finding 14 does the sum.

The works a screen wrongly rejects are a vanishing fraction of the rejected mass. At 95% screen recall, the ~3.2M screened-out records contain ~2,300 missed works: a density of **0.07%**. A sample of 600 screened-out records therefore returns an **expected 0.4 of them**, and you cannot estimate a rate from zero events. Seeing twenty would take **30,134** human codings, about **2,009 coder-hours** against the ~65 this study can buy.

So the central promise of the project (a human audit that quantifies what the pipeline missed) was, as first specified, impossible. It is recorded here rather than quietly fixed, because a preregistration that hides its own false starts is not a preregistration.

6.1 Design: three instruments, not one

(i) Score-stratified probability sampling of the screened-out set. This is salvageable, and only because the misses **concentrate**. The pilot measured where: of the 37 works screener B pulled into scope that screener A rejected, **30 sit in the contested boundary and only 6 in the settled rejects**. So:

- Partition **F** into **screened-in** (T1/T2) and **screened-out**.
- Draw a **stratified probability sample from both**, with **known, recorded selection probabilities**. Strata cross: language (EN/FR/other) × source (OpenAlex/Érudit) × partition × screener score × abstract availability.
- Oversample heavily: the **contested boundary** (where the pilot shows the misses concentrate); French records; Érudit records; adjacent STS/LIS material; **records with no abstract** (finding 11); under-represented provinces and genres.

- Sample size set to achieve a target half-width on screening sensitivity within the French stratum, the scarcest and therefore binding cell. Computed and fixed at registration.

Why stratification is binding rather than decorative: at a ~1.3% base rate a uniform sample spends its whole budget confirming obvious exclusions. Without stratification the audit is uninformative at any feasible n .

(ii) Recall against already-labelled data, not against the haystack. Any filter's recall (the topic route, the lexical route, the cheap triage in finding 13) is measured directly against the **5,737 works that already carry full-rubric labels**: run the filter over them and count what it drops. This is exactly how finding 12 scored the topic route at 12%. It costs nothing and needs no needle-hunting in the discarded mass. **No filter enters the pipeline without its recall measured this way first.**

(iii) Known-item recall on an external criterion, because (i) is blind exactly where the bias is.

Score-stratified sampling finds the works the screener *almost* caught. It is structurally blind to the ones it rejected *confidently*, and finding 11 identifies precisely which those are: T2 material (STS, LIS, the humanities) with no abstract, rejected on a title carrying none of the field's vocabulary. Such a work is not near the threshold. It is deep in the settled rejects, where (i) never looks. **An instrument that is blind where the bias lives is worse than no instrument, because it returns a confident number.**

So a third instrument, on a criterion no screener can be blind to: **venue**. A reference set of Canadian-authored articles in *Social Studies of Science*, *Scientometrics*, *Quantitative Science Studies*, *Research Integrity and Peer Review*, *JASIST*, *Research Evaluation*, *Accountability in Research*, *Recherches qualitatives*, *Documentation et bibliothèques*. **A Canadian-authored paper in Social Studies of Science is T2 by where it was published, whatever its abstract is about.**

Venue is immune to *aboutness*, which is the single thing that defeats topic retrieval (finding 12) and title-only screening (finding 11) alike. This is the frame flip applied a second time: replace “does this look like metaresearch?” with an external, checkable criterion. It is the only instrument that can see into the blind spot.

What (iii) is and is not. Known-item recall on a venue reference set is a **non-probability** estimate. It bounds and diagnoses recall on the hard cases; it does **not** replace the design-weighted population estimate from (i). Both are reported, separately, and neither is presented as the other.

6.1a A note on why agreement statistics are stratified

With a 1.3% positive rate, two screeners that both label the obvious 98% as OUT achieve high raw agreement and an unstable κ : the statistic is dominated by the cell where agreement is free. This is the well-known κ paradox with skewed marginals. Agreement is therefore reported **per stratum** and reconstructed with design weights, never read off a pooled 2×2 . The pilot already shows why this matters: agreement is 99% in the settled stratum but 95% at the boundary.

6.2 Coding

- **Two HUMAN coders, independent, blinded** to the machine labels and to each other.
- A **locked rubric** (docs/protocol/rubric.md, versioned) with worked examples and explicit edge cases.
- Disagreements adjudicated against the rubric; adjudication decisions logged.

- Agreement reported (Cohen’s κ / Krippendorff’s α) **as a property of the coding, not as evidence of classifier validity**: an agreement coefficient measures coders, not the instrument.
- **Dependency, declared**: the second coder must be bilingual (FR/EN) for the French stratum. To be recruited and **compensated** in Week 1 via CaRN, ACFAS, CAIS-ACSI, CARL. **If no francophone coder is recruited, the French-stratum estimates are withdrawn, not imputed.**

6.2a What the machine screen does, and what it cannot do

One full-frame LLM pass (a cheap model) labels every work in F against the locked rubric; a second model relabels a 20,000-record stratified sample. Disagreements are recorded and routed to the human audit. Inter-model agreement is published and **reported as a process metric**.

It is not, and will not be presented as, an accuracy estimate, and after finding 16 it is not presented as a *measurement* of anything. Agreement fails twice. First, two language models share training data, lexical priors and failure modes; their errors are correlated, and most correlated precisely on the boundary cases the field’s definition turns on. Second, finding 16: **rate agreement is not set agreement**. The cheap model matches Sonnet’s base rate within a fraction of a point and agrees with it on 98.1% of the frame, while their in-scope sets overlap 16% (10% design-weighted); and agents of one model, on one rubric and prompt, disagree beyond chance after conditioning on what each was shown, in two arms, with their ordering flipping between arms. **This is not “duplicate screening”**: that term’s warrant comes from independent *human* judgement, and borrowing it for two models would be a misrepresentation of method. The accuracy claim rests on §6.2 and nowhere else.

What the machine pass *does* contribute is real and, after finding 16, precisely bounded: **stratification** (a noisy stratifier costs the audit efficiency, never validity, because design weights use known selection probabilities), an empirically located boundary (the disagreements), and the throughput to organize 4.3M works for sampling at all. **Machine tiers are released marked provisional. Every prevalence this study reports is design-weighted human coding. No machine number is reported as the field.** The pilot’s machine base rate (§2, finding 9) stands as the motivating hypothesis the audit tests, not as a result the audit inherits.

Agent-level controls, because finding 16 and D11 demand them: agents receive randomized, fixed-size chunks recorded in an assignment manifest at run time; a duplicate-agent reliability arm re-screens a common subsample; and completeness checks verify every output file against the filesystem and fail loudly on any unreturned record, because one pilot agent reported six label files it never wrote.

6.3 Estimation

Using **design weights** (inverse selection probability): - **retrieval sensitivity within F**: the headline quantity, and the answer to “what did we miss?”; - screening **precision and recall**, with confusion-matrix counts; - **eligible count within F**, with a confidence interval; - **differential error by language, source, tradition, genre, and linkage type**: i.e. *for whom* the pipeline fails, not merely how often.

All reported with confusion-matrix counts and interval estimates, not bare percentages.

Capture–recapture is not used, and this is a considered decision, not an oversight. The routes are endogenous: R3 depends on the seed set and any author-based expansion depends on authors surfaced by R1/R2. Log-linear multi-list models cannot identify the all-zero cell without an untestable restriction on the highest-order interaction, and with endogenously constructed lists that restriction is not credible. Nor is the estimate a safe lower bound: depending on the dependence structure it may overstate the unseen population as readily as understate it. Pilot finding 7 shows the naive estimator

returning a figure that implies Canada produces the majority of the world's metaresearch. The only unconditional lower bound on the population is **the number of eligible works actually observed**, and that is what we will report. The two-phase audit replaces it because it samples the gap instead of inferring it.

7. Analysis (secondary and exploratory)

Everything in this section is **descriptive, secondary, and explicitly labelled as such**. None of it defines the field; the corpus is constructed and validated first, and only then described.

- Counts and trends by year, language, province, institution, genre, tier, and linkage type.
- Co-authorship and institutional collaboration structure (igraph; Louvain communities, betweenness, assortativity).
- Topic/thematic structure: **only** with an explicit stability assessment across seeds and preprocessing choices. Learned topics are not treated as ground truth, and if they are unstable that fact is reported rather than the prettiest run.
- Funding structure across CIHR / NSERC / SSHRC / CFI, reported with the SSHRC under-linkage of pilot finding 6 stated alongside.

8. Outputs and release

1. **Dataset** (CSV + Parquet; Frictionless schema): one row per work, carrying route provenance, the five linkage flags, language, genre, tier, classifier score, and audit status.
2. **Validation report**: design-weighted estimates, confusion matrices, and the differential bias report, **including the errors**, not only the headline accuracy.
3. **Code**: the full pipeline, targets DAG, Docker image, lockfiles.
4. **Documentation**: Quarto site, a **Datasheet for Datasets**, and this protocol with its registration stamp.
5. **A bilingual explorer** for people who do not write code.

Versioned on GitHub; each release archived to **Zenodo with a DOI**. Public corrections accepted against a transparent version history.

9. Ethics and governance

- The dataset describes **living people**. Individuals are represented only by what they have published.
- **No sensitive identity is inferred or released**. No gender, ethnicity, Indigeneity, or other protected characteristic is imputed from names, affiliations, or text.
- Where Indigenous-governed data or communities are implicated, **no derived labels are released**, and appropriate governance is sought rather than asserted. Invoking OCAP® without an Indigenous partner would be compliance theatre and this study does not do it.
- A public correction mechanism allows individuals to contest or amend records about them.

10. Reflexivity

I am an independent researcher with no institutional affiliation. I am therefore a member of the population this dataset structurally cannot see: affiliation-based country assignment fails on precisely such records, and 64% of the search space carries no affiliation string at all (pilot finding 2). The **CA-FUND** clause in the primary estimand and the **non-retrieved stratum** in the audit exist because of that failure mode.

I am also, on the primary estimand, plausibly *in* the dataset. I will not code any record in the validation sample that includes my own work, and any such record is routed to the second coder and an adjudicator.

11. Deviations

Any departure from this protocol will be recorded in `DEVIATIONS.md` with its date, its reason, and whether it was made before or after the outcome data were seen. Silent revision is itself a research-integrity failure, and this is a metaresearch project.

Appendix A. Screening rubric v2.2: LOCKED, and the instrument the funded work runs under

Status: LOCKED, 2026-07-13. Supersedes v1.0 for all screening from this date.

v2.0 → v2.1, same day, recorded rather than quietly patched. v2.0 mandated *insufficient_payload* as “a flag distinct from OUT” while *screening-schema.json*’s tier enum was [T1, T2, T3, OUT]: **v2.0 demanded a value its own output contract could not express.** That is exactly the defect v2 was written to fix (D20), committed by the document that fixes it, and the build guard aimed at that defect did not catch it because it was hardcoded to read v1’s file. A screening agent found it, in a footnote, while doing something else. The schema now carries the value; the guard now reads the current rubric and checks both directions. See *DEVIATIONS.md* D25. **The 179-work re-screen ran under v2.0**, so its three *insufficient_payload* records were coded OUT with the flag in the reason string; that is stated rather than silently corrected.

v1.0 is not deleted and is not corrected. It remains in *docs/protocol/rubric.md*, byte for byte, because 5,600 works were screened against it and every number in the pilot is attributable to that text. A rubric edited after the data are seen is a rubric that can no longer say what any past number means. See *DEVIATIONS.md* D25.

Where this definition comes from, and why that matters

v2.2 anchors the definition to the field’s own published one, and cites it. Through v2.1 the definition of metaresearch in this document was **mine**: six verbs I chose. That is an uncomfortable place to stand for a project whose central finding is that *the definition was underspecified*, and a reviewer had no way to check the instrument against anything except my judgment.

Meta-research is the study of research itself: its methods, reporting, reproducibility, evaluation, and incentives.

Ioannidis JPA. Meta-research: Why research on research matters. *PLoS Biol.* 2018;16(3):e2005468. doi:10.1371/journal.pbio.2005468 See also Ioannidis JP, Fanelli D, Dunne DD, Goodman SN. Meta-research: Evaluation and improvement of research methods and practices. *PLoS Biol.* 2015;13(10):e1002264.

Those **five domains** are now the spine of T1, and every work coded T1 also carries a domain. The definition is external, citable, and not the applicant’s invention.

Why the five domains are NOT the whole map, and why narrowing to them would be a mistake

Ioannidis’s definition is a **metascience** definition. The call for proposals **explicitly welcomes** other traditions by name: *bibliometrics and scientometrics*, *science and technology studies (STS)*, *scholarly communication research*, *open science and open scholarship research*. An STS scholar does not describe their work as “meta-research” in Ioannidis’s sense; STS has its own lineage, its own journals, and its own account of what studying science means.

So adopting the five domains **alone** would delete precisely the traditions the challenge’s inclusiveness criterion exists to protect. That is not a hypothetical risk: **OUT-vs-T2 is the single largest tier confusion in our data at every sample size**, and it is exactly this seam.

Therefore: the five domains are the T1 spine. T2 remains the adjacent traditions, named as the call names them. A work can be squarely within the field's canonical definition (T1, with a domain) or squarely within a tradition the call asks us to include (T2, without one). Both are in the map. Only the first is in the canonical corpus.

Why v2 exists, and how its sentences were chosen

Not by taste. **By census.**

Three frontier models screened 5,600 works against v1. They split on 179 of them. An independent judge, from outside the panel and blind to which model said what, adjudicated all 179 and was required to name, for every work, *which seam of the rubric produced the split*.

Its verdict on v1:

The rubric is silent on 159 of the 179 contested works (89%). Only 17 splits were a screener misapplying a rule that actually existed.

The field's boundary was being set by the screener, not by the instrument. Every model was left to invent the missing sentence privately, and none of them invented the same one.

The seams, weighted by the works they actually cost:

seam	works	% of contested
methods_dev_vs_study	39	21.8%
lis_sts_asymmetry	26	14.5%
history_of_science	19	10.6%
workforce_boundary	19	10.6%
<i>(screener error; the rubric was fine)</i>	17	9.5%
bibliometrics_contradiction	12	6.7%
instrument_validation	10	5.6%
insufficient_payload	8	4.5%
knowledge_translation	8	4.5%
guidelines_standards	7	3.9%
t3_unscoped_genre	6	3.4%
research_governance	5	2.8%
peer_review_artifact	3	1.7%

Two judges worked disjoint batches with no contact. They named the same top seam, and they independently proposed nearly the same missing sentence. That convergence is why the two rules below are stated first and stated hardest: they are not the ones I found most interesting, they are the ones that cost the most agreement.

The two rules that carry the most weight

Everything else in this document is detail. These two sentences resolve, between them, roughly **36% of every disagreement three frontier models had about this field.**

RULE 1 (resolves methods_dev_vs_study, instrument_validation, and much of guidelines_standards)

A work is T1 on methods only if its object is methods as used, reported, or performed by researchers. Developing, deriving, validating, or teaching a technique for use in a domain is OUT, however generic the technique and however rigorously its properties are established.

The discriminator, which the rubric already almost states in its error 2: *if this paper is right, what do we now know more about: a scientific question, or the practice of science?*

- A new high-breakdown estimator in a statistics journal → **OUT**. It is a contribution to research practice; it is not a *study* of it.
- Psychometric validation of a depression scale → **OUT**. Same reason. validation was flagged as a polysemy trap in v1 and never resolved; it is resolved here.
- A demonstration that feature-selection bias inflates the classification accuracies *published in a literature* → **T1**.
- A review of how 200 network meta-analyses handled inconsistency → **T1**.
- A COSMIN-style study of how a field *chooses* what to measure → **T1**.

The same confusion wears three different hats, and this one sentence removes all three: **an instance of a research practice is not a study of it**. That is v1's own error 1 ("a study that uses a method is not a study of the method"), extended to cover *building* the method as well as *using* it.

RULE 2 (resolves lis_sts_asymmetry and history_of_science)

The STS, LIS, history and philosophy of science bullets apply only where the object is science, research, or the research ecosystem. That condition governs all of them equally, at any period.

v1's LIS bullet said "information behaviour **of researchers**". The STS bullet directly above it carried no such qualifier, so STS-shaped work on non-science technology (municipal smart-city governance, public opinion on GMOs) had nothing in the text to stop it entering T2.

And "**contemporary**" is **deleted** from the history/philosophy bullet. In v1 it silently excluded the entire history of Canadian science: a history of the Canadian archival profession, the transatlantic transfer of experimental pedagogy to Quebec, how cybernetics spread through Latin America. For a project whose inclusiveness criterion is about *traditions*, deleting a whole tradition on an unexamined adverb is the worst available outcome. If a date restriction is wanted it belongs in the **estimand**, not smuggled into one tier bullet.

The unit

Unchanged from v1. One work, identified by its OpenAlex ID. You see title, abstract, year, language, venue, OpenAlex topic and field, Canadian institutional affiliations, funders.

The question

Is the object of study *research itself*?

Metaresearch studies how research is done, reported, funded, evaluated, disseminated, or governed. The test is what the work is *about*, not what field it appears in and not what vocabulary it uses.

A cardiology trial is not metaresearch. A study of *how cardiology trials report their outcomes* is.

v1 named six verbs and operationalized four. disseminated and governed had no tier bullet at all, so knowledge translation and research ethics had nowhere to land and screeners coded them OUT while saying they were unsure. Both now have homes (T1 and T2 below). A definition that promises six things and delivers four is not a strict instrument; it is an instrument with two undefended doors.

Tier

Assign exactly one. **The object decides the tier. Genre never does.** (This single principle retires `guidelines_standards` and `t3_unscoped_genre`; see T3.)

T1: Core metaresearch

The primary object of study is research itself, in one of **Ioannidis's five domains**. Record which, in domain.

domain	what it covers
methods	how research is designed, conducted and analyzed; bias; statistical practice; analytic-choice sensitivity; evidence-synthesis methodology
reporting	how research is written up and disseminated; reporting standards; scholarly communication; open access and preprints as <i>objects of study</i>
reproducibility	replication; data and code sharing; transparency; reproducibility of published findings
evaluation	peer review and editorial process; research assessment; metrics, rankings, indicators
incentives	funding allocation; the research workforce (careers, equity, training, capacity); reward structures; research integrity, misconduct and questionable research practices; research ethics and governance

The bullets below are the operational form of those five, and each names its domain.

- **research methods as practiced:** bias, reporting, statistical practice, study design, analytic-choice sensitivity — **subject to RULE 1**
- **reproducibility**, replication, data and code sharing
- **research integrity:** misconduct, retraction, questionable research practices, authorship

- **peer review** and editorial process as *objects of study* — **but a peer-review artifact whose content is a domain paper (a published referee report, an open-review record, a “Discussion of”) is an INSTANCE, not a study: OUT.** It is RULE 1 again.
- **research evaluation**, metrics, rankings, funding allocation
- **scholarly communication**: publishing models, open access, preprints as *objects of study*
- **bibliometrics and scientometrics** that study **research practice** through bibliometric means (citation distortion, reporting trends, retraction citation) — *see the bibliometrics rule below*
- the **research workforce**: careers, equity, training, capacity — **subject to the workforce rule below**
- **evidence synthesis methodology** (work *about* how syntheses are done, not an individual review’s clinical answer)
- **research ethics and governance as an object of study** — *new in v2; the definition promised “governed” and v1 delivered nothing*
- **reporting and measurement standards** (COBIDAS, core outcome sets, a Delphi harmonizing definitions so results can be compared across centres) — **T1 by object, even though a position statement by form**

T2: Adjacent

Empirically or conceptually grounded work on science as a social system that is not framed as metaresearch but studies it. **All bullets are subject to RULE 2.**

- **science and technology studies**: social studies of scientific knowledge, practice, controversy — *where the knowledge or practice studied is research*
- **library and information science**: information behaviour of researchers, repositories, discovery
- **research infrastructure** studies: platforms, data repositories, identifiers, standards — *scholarly infrastructure; a domain instrument or analysis package is OUT*
- **history and philosophy of science** where the object is science or research practice — **at any period**
- **research policy** analysis grounded in evidence
- **knowledge translation and implementation science** where the object is the **uptake of research findings** — *new in v2; health-service delivery that merely involves an intervention is OUT*
- **bibliometric or scoping studies that characterize a literature** (who publishes, where the gaps are)

T2 is where the field’s diversity lives. Be generous here: a paper does not have to call itself metaresearch to be studying research.

T3: Contextual

Relevant to the ecosystem, mapped and released, but **never pooled into the analytic corpus.**

Every T3 bullet requires the item’s OBJECT to be research or the research ecosystem, not merely its genre. A letter about a domain result is OUT. A letter about peer review is T3. An erratum on a metallurgy paper is OUT. A reviewer-acknowledgement list is T3.

- editorials, commentaries, opinion pieces **about research practice**
- policy documents and position statements **about research** (*but a reporting/measurement standard is T1, by object; see above*)

- announcements of **research-ecosystem** infrastructure, tools, or initiatives (*a bioinformatics package for a domain task is OUT*)
- news, obituaries, corrections, letters **about research or the research ecosystem**

OUT: Not meta-research

Everything else. Including, and these remain the most common errors:

- **A study that uses a method is not a study of the method.** A meta-analysis answering a clinical question is OUT. A study of how meta-analyses handle heterogeneity is T1.
- **A work that BUILDS a method is not a study of the method either.** (RULE 1. New in v2, and it is the single largest source of disagreement in the field.)
- **Mentioning a meta-research term is not studying it.** A lab paper reporting “assay reproducibility”, a methods section saying “peer reviewed”, a licence note saying “open access”: all OUT.
- **Domain research that happens to be bibliometric in method**, but whose object is the domain.

The bibliometrics rule (v1 contradicted itself here)

v1’s T1 bullet said “bibliometrics applied to a research literature”. v1’s worked example sent a bibliometric map of a domain literature to **T2**, using the T1 bullet’s own criterion as the reason. **Both could not be right, and screeners followed different ones.**

Resolved:

- A bibliometric study that **characterizes a literature** (who publishes, where the gaps are, how a field grew) → **T2**.
- A bibliometric study that **studies research practice** through bibliometric means (citation of retracted work, reporting trends over time, citation distortion) → **T1**.

The workforce rule

“The **research** workforce” was always the wording; the failure was at the boundary of the role, not in the words.

- The **research** workforce: careers, equity, training, capacity → **T1**.
- **Academics acting in a teaching role** (student evaluation of teaching, faculty development for clinician teachers), **higher-education studies** (participation policy, university writing centres), and **professional or clinical training** (nursing retention, surgical curricula) → **OUT**.

Records that cannot be screened

New in v2. Use **insufficient_payload**, a flag distinct from OUT.

The frame carries scraping artifacts as titles (178 PUBLICATIONS 2,994 CITATIONS), thesis licence boilerplate, DOAJ and Crossref persistence boilerplate in the abstract field (which is *literally text about scholarly infrastructure*, a keyword trap), records whose title and abstract belong to different papers, and outright spam.

It is not a screening judgment. It is a statement that the record cannot be screened. It must be *reported*, not silently dropped: the size of this class is a finding about OpenAlex, and the audit samples it.

A supplement (Additional file 3 of...) **inherits its parent's tier**, or is flagged insufficient_payload when the parent cannot be identified.

Canadian linkage

Split in two. v1's single boolean could not separate "Canadian data, universal claim" from "a claim about Canada", and four screening agents independently proposed exactly this repair.

- CA_AFF: at least one author affiliated with a Canadian institution. **Warning, measured:** OpenAlex invents Canadian affiliations. Impact is not an institution; it is a parse failure with an institution id, and it is attached to a Spanish COVID essay. At least 17,466 works enter the frame on an institution that does not exist (finding 27), and that is a lower bound. A Canadian affiliation is *evidence*, not proof.
- CA_FUND: funded by CIHR, NSERC, SSHRC, or CFI.
- ABOUT_CA_SYSTEM: the Canadian **research system or academy** is a substantive object (not a passing mention, not one country among forty). *A Canadian institution as **setting** is not this.* A Quebec COVID cohort is not about the Canadian research system; an analysis of Canadian law faculties' tenure norms is.
- ABOUT_CA_TOPIC: Canada is a substantive subject of the work, whether or not research is the object. (*Prairie drought, the 2006 federal election, BC grizzly bears.*)

Why this matters and is not bookkeeping: in v1, about_ca could only be TRUE once a work was *already* about research, so it was nearly collinear with tier and carried almost no independent information **in the very stratum built to carry it**. The secondary estimand could not be estimated from the stratum designed for it (finding 27).

Genre

One vocabulary. This document is the only place it is defined, and docs/protocol/screening-schema.json now QUOTES this list rather than paraphrasing it.

empirical · review · methods · commentary · editorial · protocol · dataset · software · other

In v1 the rubric and the schema named **two different vocabularies for this field, overlapping on two values out of thirteen**, and every screener was handed both and told to obey both. Two models followed the rubric and were therefore a quarter illegal against the schema; one drew from both lists at once and emitted thirteen distinct values. The validator checked tier and had never been asked to look at genre, so 16,800 labels passed every check that ran (D20, finding 24). **A codebook is code. It gets a test**, and pilot/check_instrument.R now fails the build on any contradiction between this document and the schema.

Confidence

- high: the rubric decides this cleanly.
- medium: the rubric decides, but a reasonable coder could differ.
- low: genuinely on the boundary, or the abstract is missing/too thin to judge.

The rule, stated once, here, and quoted verbatim by the schema: if the abstract is missing, judge on the title alone and set confidence to **Low unless the title is unambiguous**.

In v1 the schema *misquoted* this, saying Low was required *whenever* the abstract was missing. Since Low is what routes a record to human adjudication, and a third of the frame has no abstract, the contradiction silently changed **which records a human ever looks at** (D24).

Low is a legitimate answer and is not a failure. A screener that never says Low is not being careful; it is being confident.

Worked examples

Title (abridged)	Tier	Why
"Data sharing practices in 500 randomised trials"	T1	The object is reporting practice.
"Reproducibility of a Western blot protocol"	OUT	Assay reproducibility. Polysemy trap.
"A new high-breakdown estimator for multivariate location"	OUT	RULE 1. Builds a method; does not study one. Generic $\neq$ meta-research.
"Psychometric validation of a Persian self-injury inventory"	OUT	RULE 1. Validating an instrument for a domain construct.
"A systematic review of the measurement properties of instruments for X"	T1	Object is how the field measures, not one instrument.
"Trends in citation of retracted papers"	T1	The literature's own behaviour.
"COBIDAS: best practices for reporting neuroimaging"	T1	Object is how research should be reported. A position statement by form; the object decides .
"A published referee report on a hydrology manuscript"	OUT	RULE 1. An instance of peer review, not a study of it.
"Reviewer acknowledgement list, volume 12"	T3	Its object <i>is</i> peer review.
"Correction to: binder-jet metallurgy of Inconel 718"	OUT	T3 genre, but the object is metallurgy. The object decides .
"Laboratory life in a Canadian genomics centre: an ethnography"	T2	STS, object is research.
"Sidewalk Labs and urban socio-technical imaginaries in Toronto"	OUT	RULE 2. STS in method; the object is corporate urbanism, not science.

Title (abridged)	Tier	Why
“How cybernetics spread through Latin America, 1955-1975”	T2	RULE 2. History of science; “contemporary” is deleted.
“A history of the Canadian archival profession”	T2	RULE 2. Research ecosystem, historical.
“How do researchers discover datasets? A survey”	T2	LIS, information behaviour of researchers.
“Des services pensés pour tous” (inclusive library services)	OUT	RULE 2. LIS by venue; the object is library service users, not researchers.
“Effectiveness of statins: a network meta-analysis”	OUT	Uses a synthesis method to answer a clinical question.
“How network meta-analyses handle inconsistency: a review of 200 reviews”	T1	Object is the method.
“Bibliometric analysis of diabetes research 2000-2020”	T2	Characterizes a literature. (<i>v1’s T1 bullet said T1. The contradiction is resolved: this is T2.</i>)
“Citation distortion in the amyloid literature”	T1	Studies research practice through bibliometrics.
“Bias in student evaluation of teaching”	OUT	Academics in a teaching role. Not the research workforce.
“Career reconversion of doctoral graduates”	T1	The research workforce.
“Barriers to uptake of a clinical guideline in Ontario hospitals”	T2	Knowledge translation: object is the uptake of research. (<i>v1 had nowhere to put this.</i>)
“Announcing the Canadian Research Data Repository”	T3	Research-ecosystem infrastructure.
“OSeMOSYS Global: an open electricity-system model generator”	OUT	A tool, but for a domain task.
“Open access mandates are failing early-career researchers” (editorial)	T3	Commentary about research practice.
“178 PUBLICATIONS 2,994 CITATIONS”	insufficient_payload	A scraping artifact, not a title. Reported, not silently dropped.
“Le libre accès dans les revues québécoises : état des lieux”	T1	Scholarly communication. French.

Instructions to the screener

Read the title and abstract. Apply the rubric. Do not use outside knowledge of the authors or the journal's reputation. If the abstract is missing, judge on the title alone and set confidence to low unless the title is unambiguous.

When in doubt, ask RULE 1's question: *if this paper is right, what do we now know more about: a scientific question, or the practice of science?*

Return the schema in docs/protocol/screening-schema.json. One object per work. No prose, no preamble.

What v2 does not claim

It has not been shown to work. **It has been shown to be written against the right questions**, which is not the same thing, and the difference is the next experiment.

The 5,600-work sample will be re-screened under v2 and the **v1-to-v2 difference reported as a finding**: *what did specifying these seams actually change?* That measurement is available only because the ambiguity was enumerated, adjudicated, and published **before** it was resolved. If v2 does not move the agreement numbers, that is a result, and it will be reported as one.

Appendix B. Screening rubric v1.0: SUPERSEDED, and the instrument every pilot number was produced under

Locked before screening. Version-controlled. Any change is a new version with a dated entry in DEVIATIONS.md.

Both machine screeners and both human coders work from this document and nothing else. The rubric is the instrument; if two coders disagree, the rubric is what they are adjudicated against.

The unit

One work (journal article, preprint, book chapter, report, dataset, or software), identified by its OpenAlex ID. You see: title, abstract, publication year, language, venue, OpenAlex topic and field, Canadian institutional affiliations, funders.

The question

Is the object of study *research itself*?

Metaresearch studies how research is done, reported, funded, evaluated, disseminated, or governed. The test is what the work is *about*, not what field it appears in and not what vocabulary it uses.

A cardiology trial is not metaresearch. A study of *how cardiology trials report their outcomes* is.

Tier

Assign exactly one.

T1: Core metaresearch

The primary object of study is research itself. Typical objects:

- research **methods** and their properties (bias, reporting, statistical practice, study design)
- **reproducibility**, replication, data and code sharing
- **research integrity**: misconduct, retraction, questionable research practices, authorship
- **peer review** and editorial process *as objects of study*
- **research evaluation**, metrics, rankings, funding allocation
- **scholarly communication**: publishing models, open access, preprints *as objects of study*
- **bibliometrics and scientometrics** applied to a research literature
- the **research workforce**: careers, equity, training, capacity
- **evidence synthesis methodology** (not an individual systematic review's clinical answer, but work *about* how syntheses are done)

T2: Adjacent

Empirically or conceptually grounded work on science as a social system that is not framed as metaresearch but studies it:

- **science and technology studies (STS):** social studies of scientific knowledge, practice, controversy
- **library and information science:** information behaviour of researchers, repositories, discovery
- **research infrastructure** studies: platforms, data repositories, identifiers, standards
- **history and philosophy of science** *when the object is contemporary research practice*
- **research policy** analysis grounded in evidence

T2 is where the field's diversity lives. Be generous here: a paper does not have to call itself metaresearch to be studying research.

T3: Contextual

Relevant to the ecosystem, mapped and released, but **never pooled into the analytic corpus**:

- editorials, commentaries, opinion pieces about research practice
- policy documents and position statements
- announcements of infrastructure, tools, or initiatives
- news, obituaries, corrections, letters

OUT: Not metaresearch

Everything else. Includes, and this is the most common error:

- **A study that uses a method is not a study of the method.** A meta-analysis answering a clinical question is OUT. A study of how meta-analyses handle heterogeneity is T1.
- **Mentioning a metaresearch term is not studying it.** A lab paper reporting “assay reproducibility”, a methods section saying “peer reviewed”, a licence note saying “open access”: all OUT.
- **Domain research that happens to be bibliometric in method** but whose object is the domain, not the research: e.g. a bibliometric map of diabetes treatments, produced to summarise diabetes knowledge. Borderline; see below.

The three errors that matter most

These are where the pilot showed retrieval and screening break down. Read them twice.

1. **Polysemy.** reproducibility in a biochemistry paper means the assay gives the same reading twice. In a metaresearch paper it means the reproducibility crisis. Only the second is in scope. The same applies to validation, replication, bias, peer review, open access. **Judge the sense, not the token.**
 2. **Method vs. object.** Using bibliometrics $\neq$ studying research. Ask: *if this paper is right, what do we now know more about: a scientific question, or the practice of science?* If the former, OUT.
 3. **Absence of vocabulary is not absence of content.** An STS paper about laboratory practice, or an LIS paper about how researchers find data, may use none of the field's keywords and still be squarely T2. **Do not require the vocabulary.** This is the error that makes the field look smaller and more anglophone than it is.
-

Canadian linkage

Record which apply (this is metadata, not an inclusion test; the estimand is defined separately in PROTOCOL.md):

- CA_AFF: at least one author affiliated with a Canadian institution
- CA_FUND: funded by CIHR, NSERC, SSHRC, or CFI
- ABOUT_CA: the Canadian research system is a substantive object of study (not a passing mention, not one country among forty in a global comparison)

Genre

empirical · conceptual · editorial/commentary · policy · infrastructure/announcement · other

OpenAlex's type is unreliable and this field is commentary-heavy, so genre is coded, not inherited.

Language of the work

As published. Do not infer from the venue.

Confidence

- high: the rubric decides this cleanly.
- medium: the rubric decides, but a reasonable coder could differ.
- low: genuinely on the boundary, or the abstract is missing/too thin to judge.

Low is a legitimate answer and is not a failure. Records coded low by either screener are routed to adjudication. A screener that never says low is not being careful; it is being confident.

Worked examples

Title (abridged)	Tier	Why
"Data sharing practices in 500 randomised trials"	T1	The object is reporting practice.
"Reproducibility of a Western blot protocol"	OUT	Assay reproducibility. Polysemy trap.
"Trends in citation of retracted papers"	T1	Object is the literature's own behaviour.
"Laboratory life in a Canadian genomics centre: an ethnography"	T2	STS. No metaresearch vocabulary; still studies research.
"How do researchers discover datasets? A survey"	T2	LIS, information behaviour of researchers.

Title (abridged)	Tier	Why
“Effectiveness of statins: a network meta-analysis”	OUT	Uses a synthesis method to answer a clinical question.
“How network meta-analyses handle inconsistency: a review of 200 reviews”	T1	Object is the method.
“Announcing the Canadian Research Data Repository”	T3	Infrastructure announcement. Mapped, not pooled.
“Open access mandates are failing early-career researchers” (editorial)	T3	Commentary about research practice.
“Bibliometric analysis of diabetes research 2000–2020”	T2	Borderline. Bibliometric method, and the object <i>is</i> a research literature, so it studies research output. Code T2, medium.
“Le libre accès dans les revues québécoises : état des lieux”	T1	Scholarly communication, object is publishing practice. French.

Instructions to the screener

Read the title and abstract. Apply the rubric. Do not use outside knowledge of the authors or the journal’s reputation. If the abstract is missing, judge on the title alone and set confidence to low unless the title is unambiguous.

Return the schema in docs/protocol/screening-schema.json. One object per work. No prose, no preamble.

Appendix C. The seam census: how v2 was derived, published before it was resolved

Status: PROPOSED. Not locked, not applied. The v1 rubric (docs/protocol/rubric.md) is the instrument the pilot ran under, and changing it after seeing the data would be fitting the instrument to the data. This document is the **evidence for a revision**, to be locked *before* the full screen and recorded in DEVIATIONS.md as a versioned change.

Where this evidence comes from, and why it is unusual

Fifteen Opus screening agents classified 5,000 works from the real frame against the locked v1 rubric, in two cohorts (five agents on works 1 to 2,000; ten agents on works 2,001 to 5,000). They worked **independently**, on **disjoint chunks**, with **no knowledge of each other**, and were asked, at the end, which boundary cases recurred.

They converged. Not loosely: agents seeing entirely different works independently reported the *same* underspecified distinctions, often naming the same axis in the same terms, and the second cohort reproduced the first cohort's seams without having seen them. That convergence is what makes this evidence rather than opinion. A distinction that fifteen frontier-model screeners, reading one rubric, cannot apply consistently **is not a distinction. It is a wish**, and the disagreement it produces is not noise to be averaged away: it is the field's boundary showing you where it actually is.

Finding 22 measured the consequence at the tier level. Finding 24 measured something worse: **the instrument contradicts itself**, and the screeners found that too.

Two facts about v1 that are not opinions

Before the seams, two defects that can be checked against the text rather than argued about.

A. The rubric and the schema name different vocabularies for genre. docs/protocol/rubric.md says empirical · conceptual · editorial/commentary · policy · infrastructure/announcement · other. docs/protocol/screening-schema.json says empirical · review · methods · commentary · editorial · protocol · dataset · software · other. **They overlap on two values out of thirteen.** Every screener was handed both and told to obey both; each invented a private reconciliation, and they did not invent the same one. Four separate agents reported this unprompted. See finding 24 and DEVIATIONS.md D20. It is now quarantined in docs/protocol/known-defects.json and make lint fails on any *further* contradiction between the two documents.

B. Two of the six verbs in the rubric's own definition have no tier bullet. The definition says meta-research studies how research is "*done, reported, funded, evaluated, disseminated, or governed.*" Four of those verbs map onto T1 bullets. Two do not:

- **disseminated:** scholarly communication covers publishing models, open access and preprints, which is dissemination of *outputs into the literature*. Nothing covers **knowledge translation and implementation science**, i.e. how research reaches practice. Four agents flagged it, and all four coded such work OUT while saying they were unsure.

- **governed:** nothing in T1 covers **research ethics, human-subjects governance, or research-governance institutions**, though governed is the promise the definition makes.

A definition that names six objects and operationalizes four is not a strict instrument. It is an instrument with two undefended doors.

The seams, ranked by how many independent screeners raised them

1. Domain methodology vs. research methods, and its sharper form: *developing* a method vs. *studying* one (raised by 15 of 15)

The single largest source of disagreement, named by every screener in both cohorts.

T1 says “research methods and their properties (bias, reporting, statistical practice, study design)”. Read one way that covers every methods paper in every field: a SNP-caller benchmark, a neuroimaging pipeline, a photoionization model, a thyroid-nodule segmentation network. Read another way it covers only cross-cutting research practice.

The second cohort sharpened it into a distinction the first cohort had not isolated. **The bullet never says whether proposing a new method counts, or only studying methods as they are used across a literature.** A new high-breakdown multivariate estimator in the *Canadian Journal of Statistics* is T1 by the letter and OUT by the spirit; every worked example in the rubric is a study of a literature, so nothing anchors the line. Screeners independently converged on the same rule and independently said they could not defend it from the text:

“results depend on the analytic choice” studies → T1; building a tool for a domain task → OUT.

A paper showing feature-selection bias inflates published classification accuracy is T1. A paper applying a new model to patient satisfaction is OUT. But a paper showing that a standard oceanographic estimator underestimates respiration threefold *studies a method’s properties across a literature* and its payoff is a domain claim, and nothing in v1 decides it.

Proposed rule. *T1 covers work whose object is research practice across a literature (reporting, bias, synthesis methodology, statistical practice, analytic-choice sensitivity), including method-comparison and method-critique studies whose finding generalizes to how research is done. It does not cover development of a technique for a domain task, nor the introduction of a new method as such, unless the work’s contribution is a claim about research practice. The discriminator, stated as the rubric’s error 2 already almost states it: if this paper is right, what do we now know more about: a scientific question, or the practice of science? Add both worked examples, and add a new-method example explicitly.*

2. Measurement and psychometrics (15 of 15)

Instrument-validation studies recur constantly: a MoCA norms paper, a TAS-20 refinement, a French-Quebec CCC-2 adaptation, a systematic review of minimal important differences for patient-reported outcomes, ActiGraph step-count validity, ICD-10 code validation, external validation of six published prediction equations.

Is validating a measurement instrument “research methods and their properties” (T1), or is it domain research whose object is a clinical construct (OUT)? The rubric flags validation as a polysemy trap *without saying how to resolve it*, which is the worst of both worlds: it tells the screener the word is dangerous and then leaves the danger.

Every screener invented a line, and **they invented compatible but unstated lines**: single-instrument validation in a domain → OUT; methodological work about *how instruments get built or should be chosen* (COSMIN-style, core outcome sets, cross-instrument reviews) → T1. One agent noted the hardest case: a validity review of a device *explicitly justified by its use as a research reference standard*.

Proposed rule. *Validation of a specific instrument for a domain construct is **OUT**, even when the instrument is used in research. Work whose object is instrument-development methodology, cross-instrument comparison, core outcome set methodology, or how a field chooses what to measure is **T1**. State it, with both worked examples, and resolve validation explicitly in the polysemy section.*

3. T3’s document-type bullets are unscoped (13 of 15)

T3 lists “news, obituaries, corrections, letters” and “announcements of infrastructure, tools, or initiatives” **without saying the item must be about research**. Only the *editorials* bullet carries the qualifier “about research practice”. Taken literally, T3 sweeps in every clinical case letter, every domain erratum, and every bioinformatics software release in existence.

Screeners split, and said so: a reviewer-acknowledgement list is T3 (its object is peer review); an erratum on a binder-jet metallurgy paper is OUT (its object is metallurgy). Under the literal reading both are T3, and T3 fills with domain corrections.

Proposed rule. *Every T3 bullet requires the item’s object to be research or the research ecosystem, not merely its genre. A letter about a domain result is OUT. A letter about peer review is T3. T3 infrastructure means **research-ecosystem** infrastructure: domain analysis software and scientific instruments are OUT. Move the “about research practice” qualifier from the editorials bullet up into T3’s heading, where it governs all four.*

4. Peer review performed is not peer review studied (5 of 15, and entirely new in cohort 2)

T1 covers peer review “as objects of study”. The frame turns out to contain peer review **as an artifact**: published referee reports, open peer-review records, PREReviews of preprints, “Discussion of” comments. OpenAlex even has a peer-review record type.

These are *instances* of the review system whose content is dogwhelks, or hydraulic modeling, or a bacteriology preprint. The rubric never says whether the review system’s own outputs are in the map. Screeners split them from reviewer-acknowledgement lists on instinct.

Proposed rule. *A peer-review artifact whose content is a domain paper is **OUT**; it is an instance, not a study, and this is the same rule as error 1 (a study that uses a method is not a study of the method). A peer-review artifact whose object is the review process itself is **T3**. Say so, because the frame is full of them and they are currently coded by coin-flip.*

5. History of science falls into a hole the rubric dug (7 of 15)

T2 admits history and philosophy of science **only** “when the object is contemporary research practice”. The STS bullet immediately above it carries **no such qualifier**. So a historical social study of scientific knowledge satisfies one bullet and fails the other, and the screeners split on exactly that.

The consequence is not a rounding error. Coded strictly, **the history of Canadian science largely drops out of the map**: the politicization of Ayurveda around Indian independence, how cybernetics spread through Latin America, how the Université de Montréal built its psychiatry training from a 1962 survey of US programs, Penfield and Jasper, a history of the Canadian archival profession. For a project whose inclusiveness criterion is about *traditions*, silently excluding an entire tradition on an unexamined adverb is the worst available outcome.

***Proposed rule. Delete “contemporary”.** Qualify the STS and HPS bullets identically: the object must be **science, research, or the research ecosystem**, at any period. If a date restriction is wanted it belongs in the estimand, not smuggled into one tier bullet where it silently deletes a tradition.*

6. LIS and STS are scoped asymmetrically (6 of 15)

T2’s LIS bullet says “information behaviour **of researchers**”. The STS bullet carries **no equivalent qualifier**, which makes STS-styled work on non-science technology (network neutrality, Sidewalk Labs’ urban socio-technical imaginaries) genuinely undecidable, and lets general library-service evaluation drift in under “be generous”.

***Proposed rule.** Qualify both bullets the same way: the object must be **science, research, or the research ecosystem**. This is the seam that produces the OUT-vs-T2 confusion finding 22 measured, so it matters most for the inclusiveness criterion.*

7. Bibliometrics: T1 and the worked example flatly contradict each other (6 of 15)

T1 lists “bibliometrics and scientometrics applied to a research literature”. The worked example sends “Bibliometric analysis of diabetes research 2000-2020” to **T2**, with the reasoning “*the object is a research literature*”. **That is the T1 bullet’s own criterion, used to justify T2**. Both cannot be right, and screeners followed different ones.

***Proposed rule.** Pick one and delete the other. Recommended: a bibliometric/scoping study that characterizes a literature (who publishes, where the gaps are) is **T2**; one that studies research practice through bibliometric means (citation distortion, reporting trends, retraction citation) is **T1**. Then fix the worked example so it stops contradicting the bullet.*

8. The research workforce: the rubric says “research”, the screeners still could not apply it (8 of 15)

Correction to an earlier draft of this document, which claimed the rubric fails to say “research workforce”. It says it. The bullet reads “the **research workforce**: careers, equity, training, capacity”. The seam is real but it is not the one I first wrote down, and it is worth being precise about, because misdescribing your own instrument is how you come to fix the wrong thing.

The actual failure is at the **boundary of the role**, not the wording:

- Academics whose role spans **teaching and research** have no home: bias in student evaluation of teaching, faculty development for clinician teachers, a protocol for developing nursing academic leaders who lead both.
- **Higher-education studies** generally: participation policy, university writing centres, spending and student engagement across 18 Ontario universities.
- **Professional and clinical training**: nursing retention, medical-student debt, surgical curricula, interprofessional education. These are professional populations, not the research workforce, and screeners flagged them as the most common near-miss after the polysemy trap.

***Proposed rule.** Keep “research workforce” and add the exclusion explicitly: the professional (non-research) workforce, and classroom or clinical pedagogy, are OUT, including when the population is academics acting in a teaching role. Add a worked example on each side of that line, because it is a line, not a wish, and it can be drawn.*

9. Records with no judgeable content (6 of 15)

The frame carries bare-DOI titles, .mat filenames, “Additional file 3 of...”, GBIF occurrence downloads, Crossref persistence boilerplate (which is *literally text about persistent identifiers*, a scholarly-infrastructure keyword trap), DOAJ open-access boilerplate in the abstract field (a false metaresearch signal), an SEO spam page, a hockey podcast, a woodworking magazine. One agent found a record whose **title and abstract are about different papers** (a body-image title with a bacterial-injectisome abstract).

These are all coded OUT/low today, which **inflates the low-confidence rate and pollutes the adjudication queue** with records that have nothing to adjudicate.

***Proposed rule.** Add an **insufficient_payload** flag, distinct from OUT. It is not a screening judgment; it is a statement that the record cannot be screened. It must be reported, not silently dropped: the size of this class is a finding about OpenAlex, and the audit samples it.*

10. Knowledge translation and research governance are promised and never delivered (5 of 15)

See fact **B** above. The definition says research is “disseminated” and “governed”; no tier bullet covers knowledge translation, implementation science, or research ethics and governance. Screeners consistently coded implementation work OUT while saying the preamble seemed to want it in.

***Proposed rule.** Decide, in the text, and say which way. Recommended: **knowledge translation and implementation science, where the object is the uptake of research findings, is T2**; health-service delivery that happens to involve an intervention is OUT. **Research ethics and governance as an object of study is T1**. Either add the bullets or remove the verbs from the definition, but do not keep promising six things and operationalizing four.*

11. Guidelines, consensus standards, and reporting standards (4 of 15)

The COBIDAS neuroimaging best-practice report, a Delphi harmonizing surgical definitions *explicitly so results can be compared across centres*, a core-outcome-set consensus statement. These are **reporting-standard work by motivation** and **position statements by form**, so T1’s methods bullet and T3’s “position statements” bullet both claim them.

Proposed rule. A consensus standard whose object is **how research should be reported, measured, or compared** is **T1**, regardless of its genre being a position statement. A position statement about policy is **T3**. Genre does not decide tier; the object does. (This is the same principle as seam 3 and it should be stated once, globally.)

12. Supplements should inherit their parent (3 of 15)

“Additional file N of [a scoping review of airway registries]” has no content of its own. Screeners judged the implied parent, which is sensible and is **nowhere in the rubric**.

Proposed rule. State it: a supplement inherits its parent’s tier, or is flagged insufficient_payload when the parent cannot be identified.

13. about_ca is near-degenerate, and it carries the secondary estimand (4 of 15, and measured)

about_ca is defined as “the Canadian research system is a **substantive object of study**”. Three problems, and the third is measured, not argued.

- T3 items are **not studies**, so the definition does not literally apply to them, and screeners diverged on an NRC audit plan and a *Canadian Historical Review* editorial.
- **A Canadian institution as setting is not about_ca**. A Quebec COVID cohort is not about the Canadian research system; an analysis of Canadian law faculties’ tenure norms is. Screeners applied this correctly but it is nowhere written.
- **It fires on about 1% of the sample**. In the about_only stratum, whose *entire retrieval basis* is that Canada appears in the title or abstract, screeners marked about_ca true for **1.8% (Opus)**, **1.3% (GPT-5.6)**, **0.5% (Grok)**. The route retrieves “Canada is mentioned”; the rubric asks for “the Canadian research system is studied”. These are different questions, and the gap between them is the polysemy finding again, applied to the country term.

The consequence is structural: **the secondary estimand of this study (“metaresearch about the Canadian research system”) currently rests on a few dozen sampled events, and the three models disagree about it by more than they disagree about tier.**

Proposed rule. Extend the definition to non-study genres (“substantive object”, not “object of study”); add the setting-vs-object exclusion with both worked examples. And, in the protocol rather than the rubric: the audit must **oversample about_ca candidates**, because at a 1% hit rate the secondary estimand cannot be estimated from a sample designed for the primary one.

Two further polysemy traps, found in the wild

v1 names reproducibility, peer review, open access, open science, validation, replication, bias. The screeners hit more, and hit the named ones in unnamed senses:

- **validation / evaluation**: institutional policies on *student* assessment, clinical *instrument* validation, health-professions competence assessment. Named as a trap, never resolved (seam 2).
- **audit**: financial audit, audit fees, clinical audit.

- Plus the observed traps: "peer reviewed: yes" as a repository artifact; judicial *standard of review*; prison *misconduct*; managerial *reporting*; DOAJ and Crossref boilerplate that is *literally about scholarly infrastructure*; and, memorably, "**Canada dry**" as a **French idiom**.
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What this does NOT license

This document does not change the pilot's numbers, and must not. Every figure in the proposal was produced under v1, and re-screening under v2 to get a better number would be exactly the failure this project exists to object to.

The correct sequence, and the one the protocol commits to:

1. Lock v2 **before** the full screen, with this evidence attached.
2. Record the change in DEVIATIONS.md as a versioned rubric revision, and **delete the quarantine entry** in docs/protocol/known-defects.json so make lint enforces instrument self-consistency from that point on.
3. Re-screen the 5,000-work sample under v2 and **report the difference against v1 as a finding**: *what did specifying these thirteen seams actually change?* That is a measurement nobody normally makes, and it is available only because the ambiguity was documented before it was resolved.